

Bachelor of Technology — General Engineering - Fintech

Table 1: General Engineering - Fintech — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	32
	Discipline Electives	34
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech General Engineering - Fintech — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-201	Data Handling and Visualisation	2-0-2-3	3
2	EE-261	Electrical Systems Around Us	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	CS-671	Deep Learning and Applications	3-1-0-4	4
5	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
6	DS-412	Matrix Computations for Data Science	3-0-2-4	4
7	IC-253	Programming and Data Structures	2-0-2-3	3
8	MA-546	Introduction to Quantitative Finance		4
9	EE-302	Control Systems	3-0-2-4	4
Total				32

Table 3: B. Tech General Engineering - Fintech — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech General Engineering - Fintech — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech General Engineering - Fintech — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-201	Data Handling and Visualisation	2-0-2-3	3
4	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
6	DE	EE-203	Network Theory	3-0-0-3	3
7	DE	IC-241	Material Science for Engineers	3-0-0-3	3
8	DE	ME-212	Product Manufacturing Technology	2-0-3-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech General Engineering - Fintech — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CS-671	Deep Learning and Applications	3-1-0-4	4
3	DC	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
4	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
5	DC	IC-253	Programming and Data Structures	2-0-2-3	3
6	DC	MA-546	Introduction to Quantitative Finance		4
7	DE	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech General Engineering - Fintech — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-302	Control Systems	3-0-2-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech General Engineering - Fintech — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech General Engineering - Fintech — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech General Engineering - Fintech — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 34 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: General Engineering - Fintech — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-313	Computer Networks	4
2	DS-312	Computing Systems for Data Processing	4
3	EE-530	Optimization theory	3
4	MA-353	Modelling of infectious disease	3
5	MA-356	Modelling and Simulation	3
6	MA-460	Nonlinear Dynamics and Chaos	3
7	MA-548	Abstract Algebra	4
8	MA-550	Statistical Data Analysis	3
9	MA-552	Number Theory	3
10	MA-553	Mathematical Foundations of Financial Engineering	3
11	MA-560	Nonlinear Dynamics and Chaos	3

General Engineering - Fintech — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	MA-563	Statistical Computation and Simulation	4
13	MA-565	Numerical Methods in Quantitative Finance	3
14	MA-572	Applied Multivariate Statistical Analysis	4
15	MA-604	Introduction To Game Theory	3
16	MA-608	Computational Fluid Dynamics	3
17	MA-641	Operations Research	3
18	MA-644	Dynamical Systems	3
19	MA-650	Mathematical Models for Infectious Diseases	3
20	MA-651	Optimization Techniques	3
21	MA-652	Stability Theory of Differential Equations	3
22	MA-653	Computational Financial Modelling	3
23	MA-654	Financial Engineering	3
24	MA-655	Fixed Income Securities	3
25	MA-656	Stochastic Calculus for Financial Engineering	3
26	MA-665	Semigroup Of Bounded Linear Operators	3
<i>Compulsory project components — counted as DE, not selectable</i>			
27	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
28	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4